

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

- To solve this , I need to use de Broglie wavelength formula $\lambda = h / \sqrt{2 * m * E}$, where h is Planck's constant , m is electron mass , and E is energy .

- Use $E = 1 \text{ eV}$, convert eV to joules ($1 \text{ eV} = 1.6 * 10^{-19} \text{ J}$) and solve . Alternatively , you can use the approximate relationship for non-relativistic kinetic energy ($1 \text{ eV} = 1240 \text{ nm}$, but a less direct relation h/p is required).

- Use the given wavelength conversion where h is about $4.14 * 10^{-15} \text{ eV-s}$ and the electron mass $m = 9.109 * 10^{-31} \text{ kg}$. Simplifying using these and check the closest match to wavelength .

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<answer> $1.23 * 10^{-7} \text{ cm}$ </answer>

